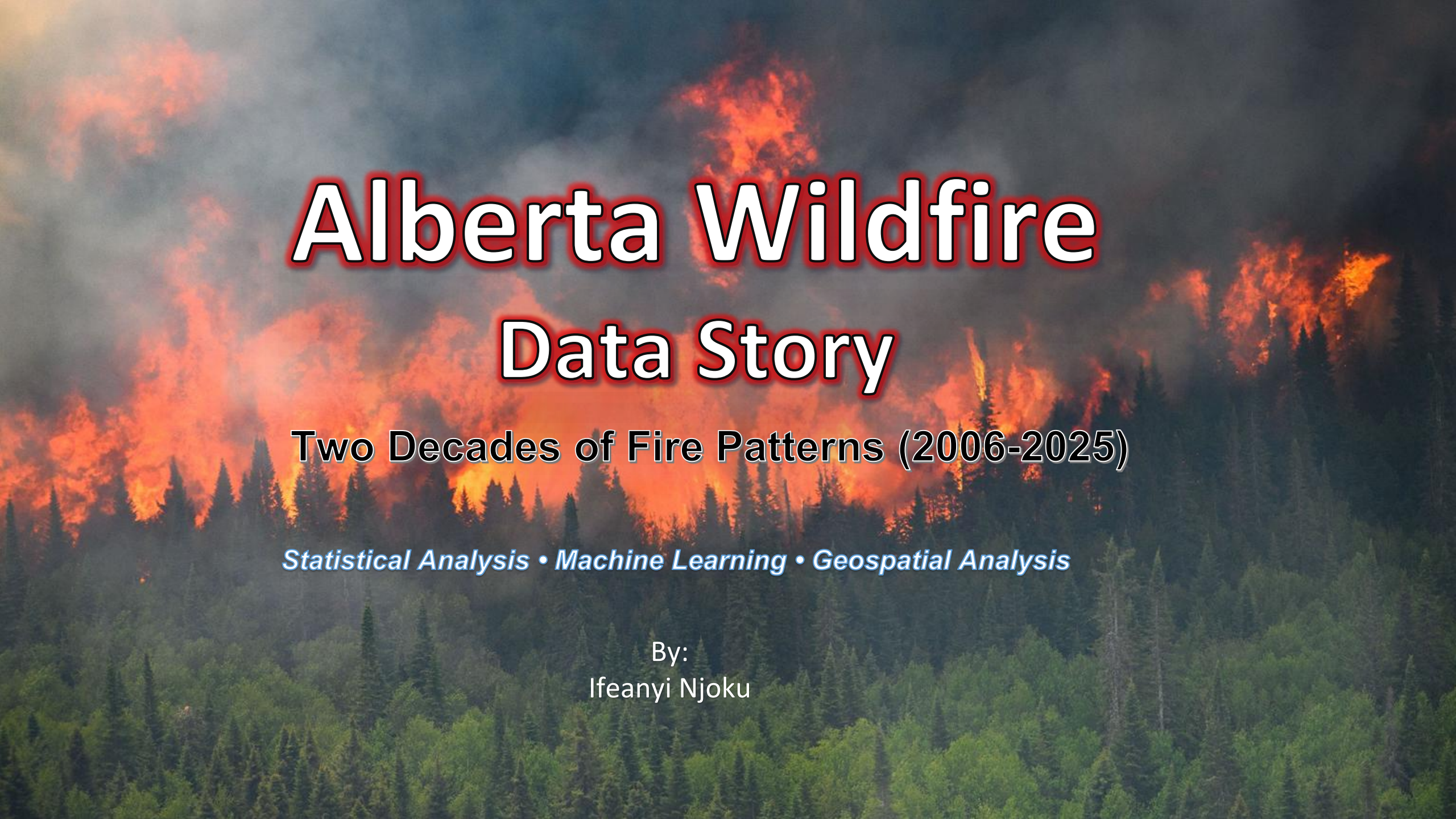




Alberta Wildfire Data Story

Two Decades of Fire Patterns (2006-2025)

Statistical Analysis • Machine Learning • Geospatial Analysis

By:
Ifeanyi Njoku

The Challenge

Every spring, Alberta faces a critical question:

- Where will wildfires strike?
- And how severe will they be?

2016 FORT MCMURRAY

88,000 evacuated

590,000 ha burned

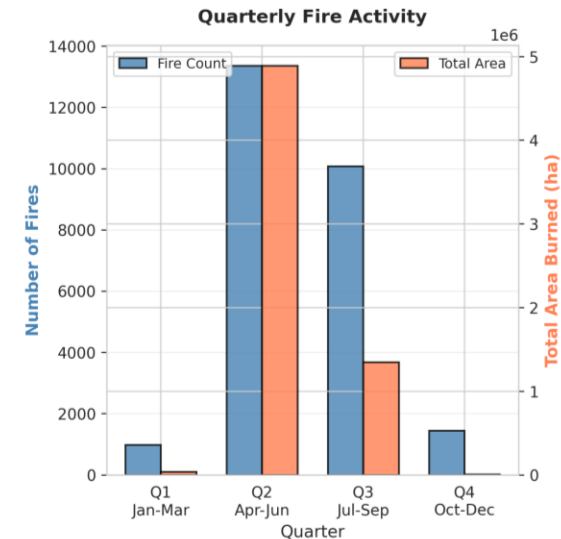
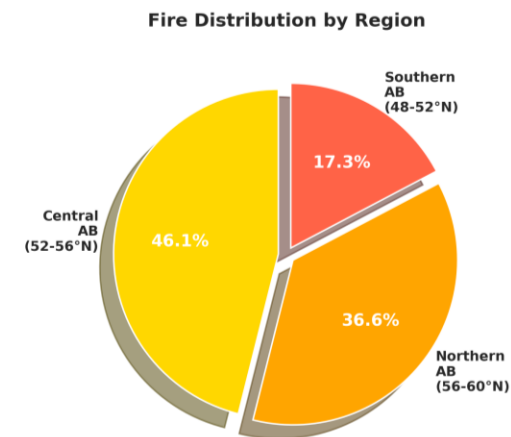
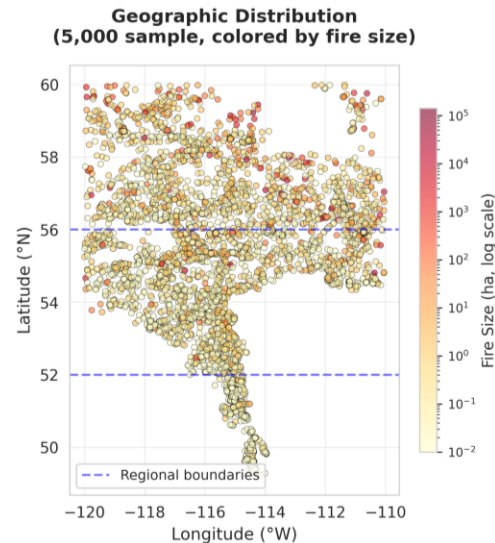
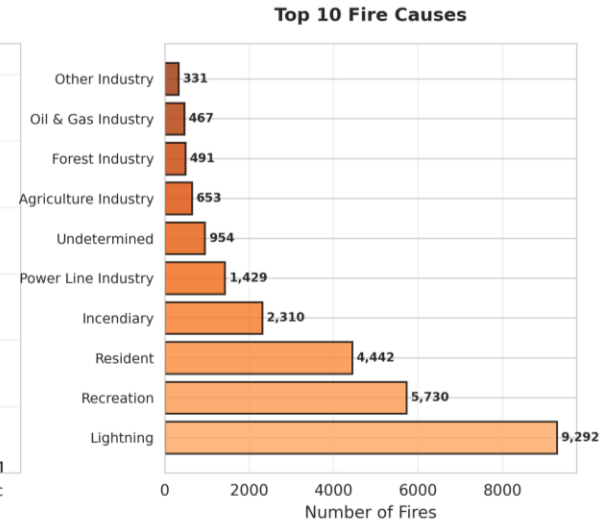
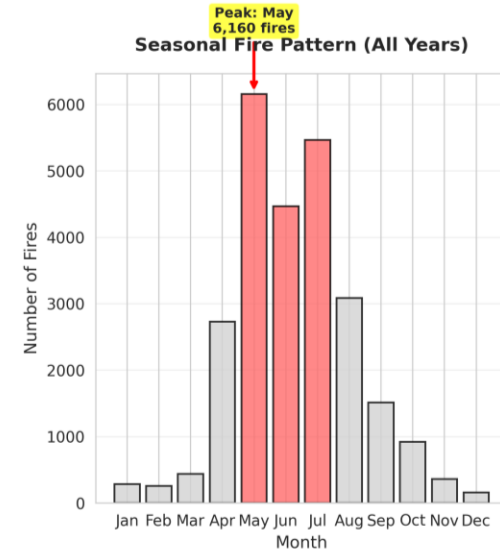
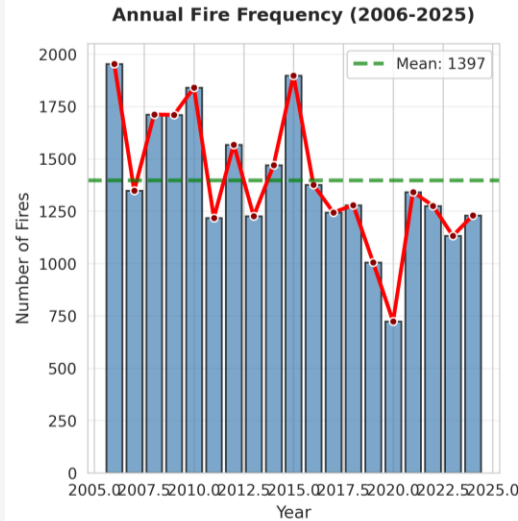
Canada's costliest natural disaster

2023 FIRE SEASON

2.7 million ha burned

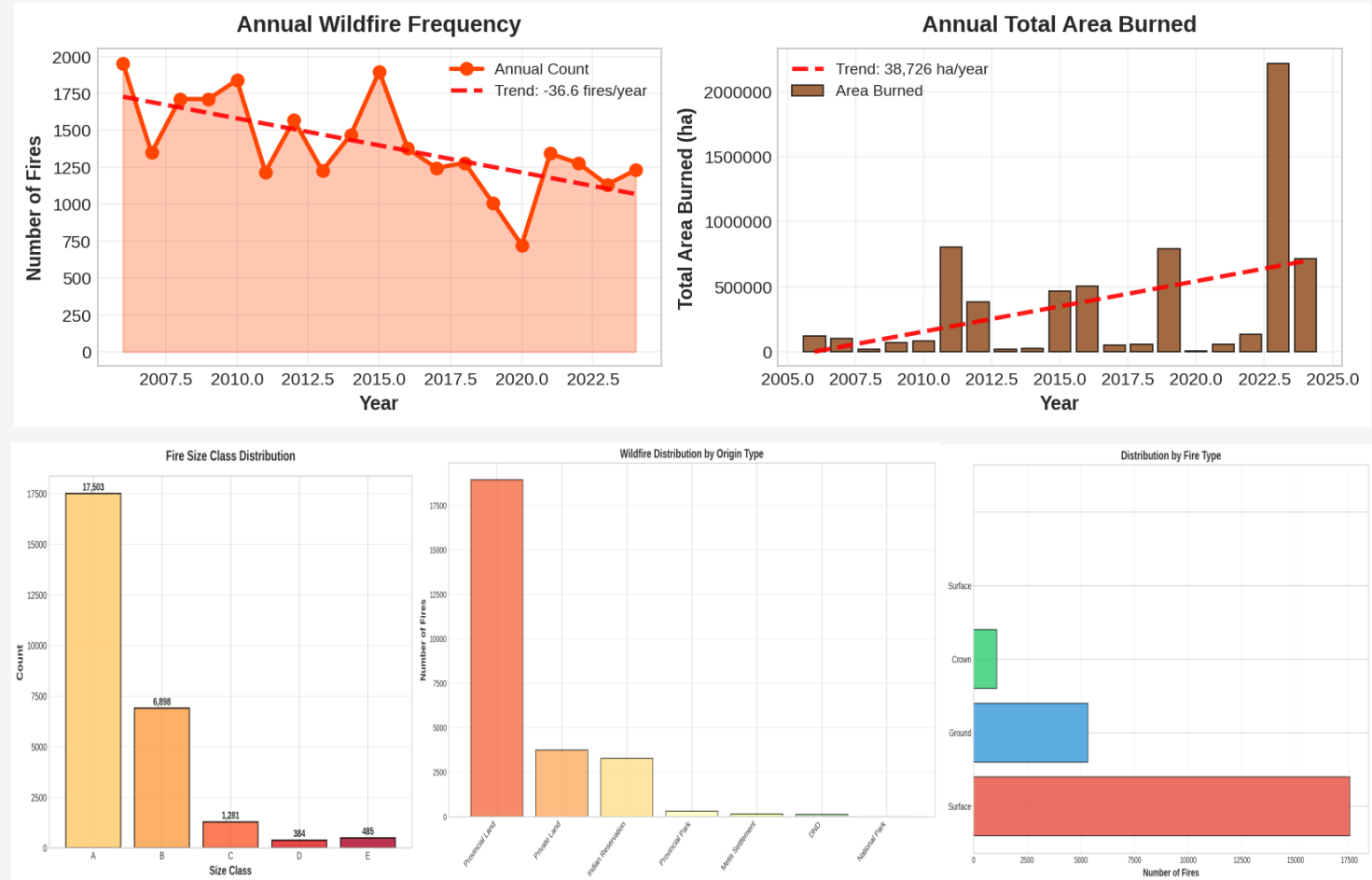
Worst on record

Air quality impacted across North America



Six Research Questions

- 1 Are wildfires increasing in frequency or severity?
- 2 Where are fires spatially concentrated?
- 3 What ignites fires, and does it vary by region?
- 4 Does faster detection reduce fire size?
- 5 Which weather factors predict fire behavior?
- 6 Can machine learning identify fire types?



Q1: Temporal Trends

Are wildfires increasing over time?

FINDING

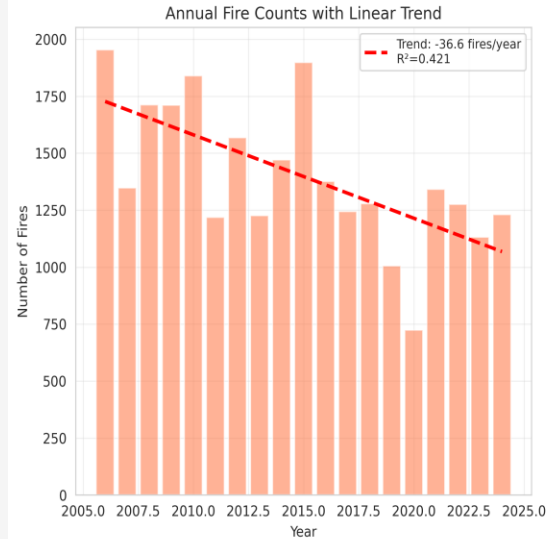
No consistent linear trend detected

METHODS:

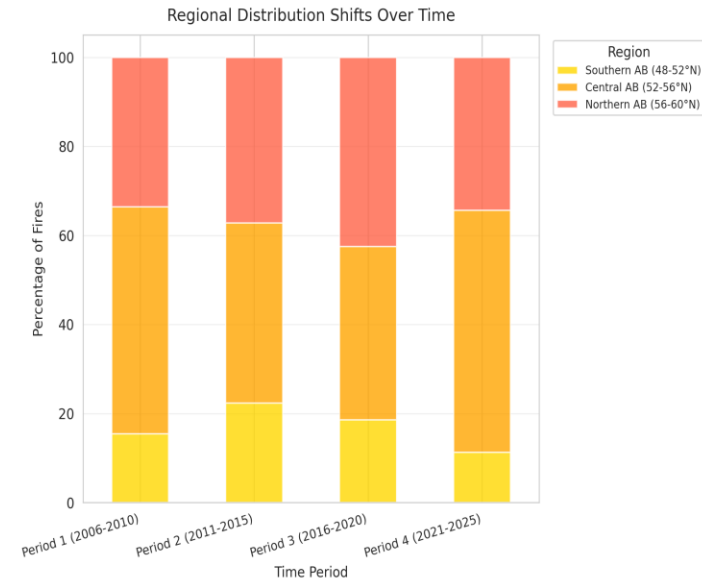
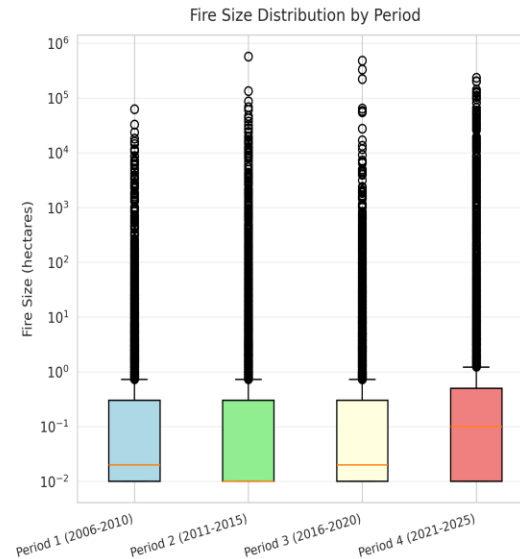
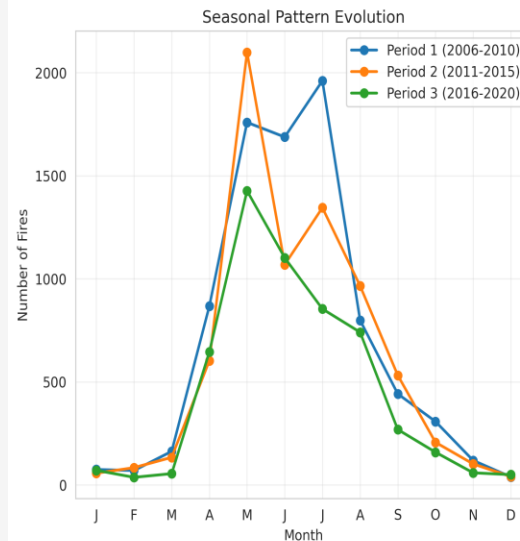
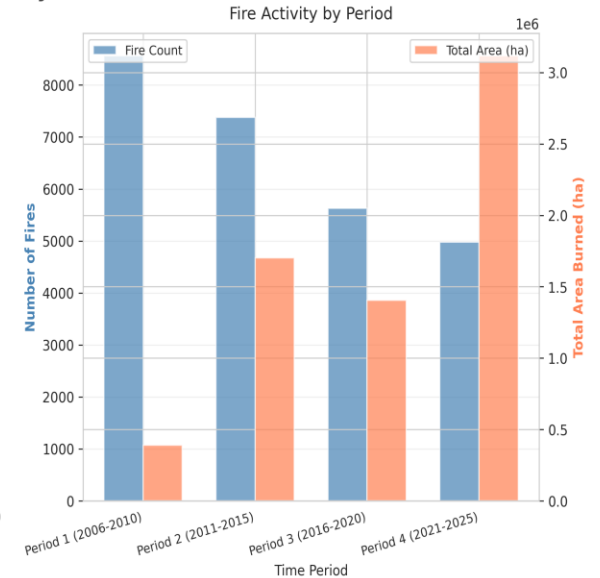
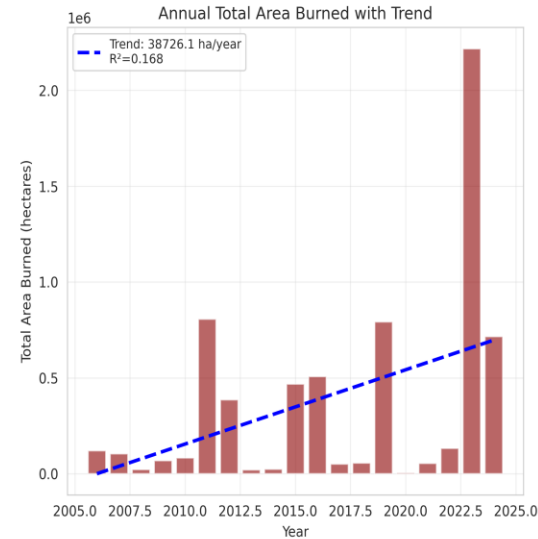
- Linear regression analysis
- Period-by-period comparison (4 periods)
- Seasonal decomposition
- Pearson correlation testing

IMPLICATIONS:

- Plan for surge capacity, not steady growth
- Extreme years cluster (2016, 2019, 2023)
- Weather drives variation more than time
- Need contingency plans for peak years



Question 1: Detailed Temporal Change Analysis



Q2: Spatial Distribution

Where are fires concentrated?

NORTHERN ALBERTA (>56°N)

- Remote boreal forest
- Lightning-caused (82%)
- Large fires
- Helicopter access needed

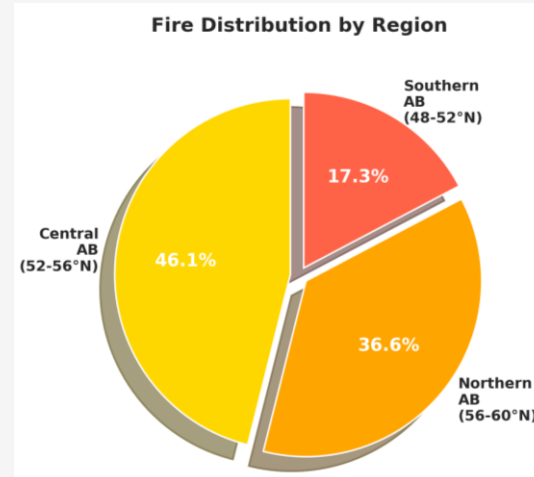
CENTRAL ALBERTA (52-56°N)

- Transition zone
- Mixed causes
- Moderate size
- Flexible strategies

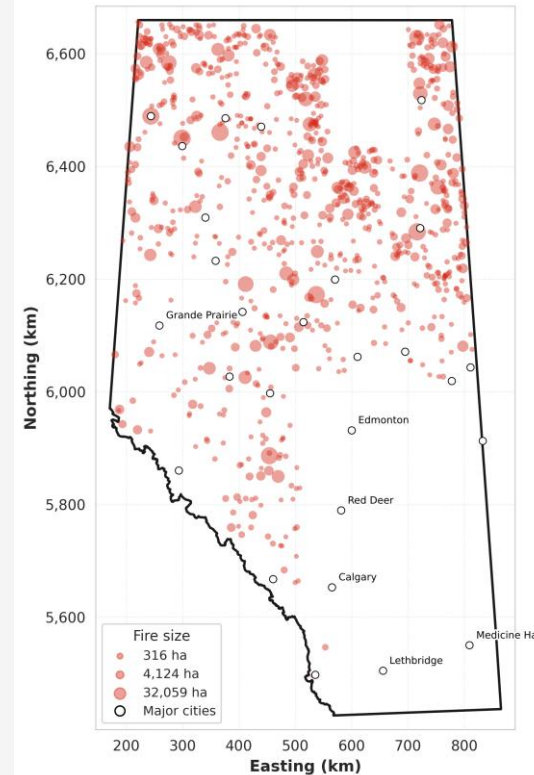
SOUTHERN ALBERTA (<52°N)

- Grassland fires
- Human-caused (51%)
- Smaller fires
- Near communities

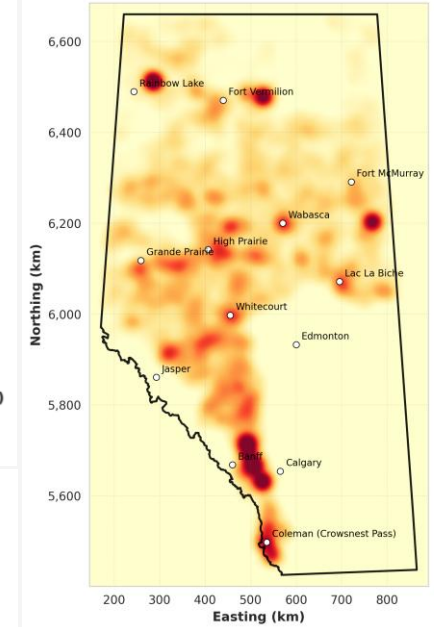
Three distinct fire environments require different management strategies



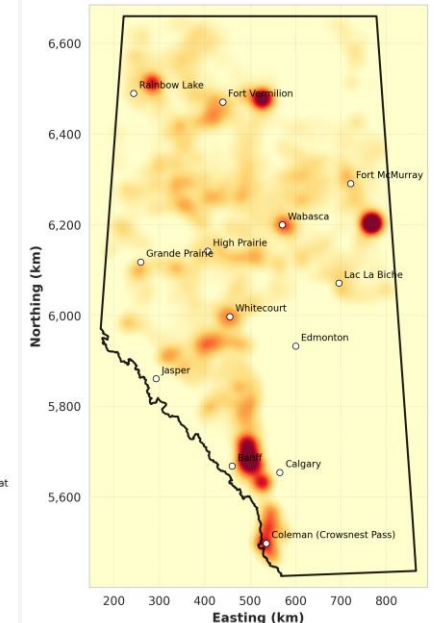
Large Fire Locations (Size-Proportional)
2006-2025 | Size Class D/E



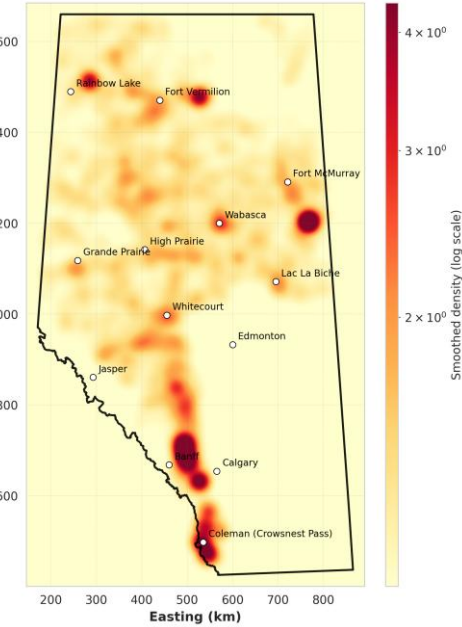
2006-2010 | Wildfire Hotspot Density
Records plotted: 8,564



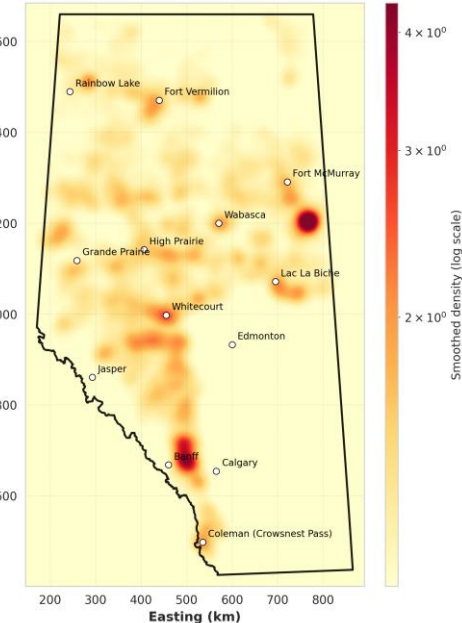
2016-2020 | Wildfire Hotspot Density
Records plotted: 5,627



2011-2015 | Wildfire Hotspot Density
Records plotted: 7,380

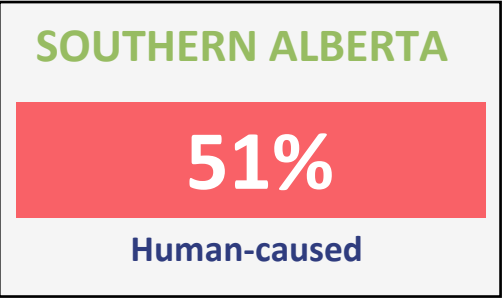
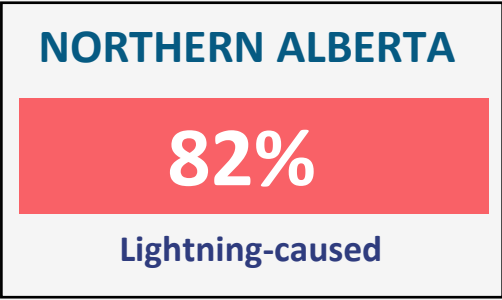


2021-2025 | Wildfire Hotspot Density
Records plotted: 4,980



Q3: Fire Causes by Region

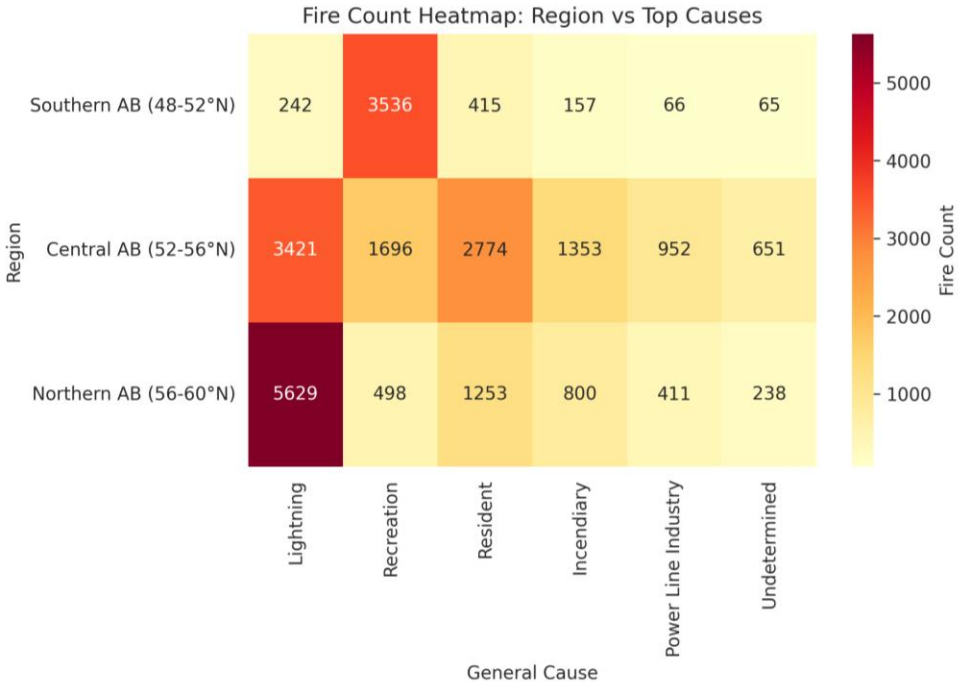
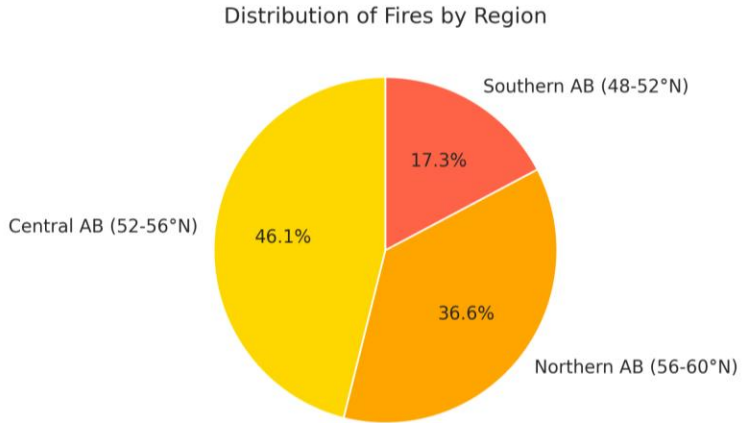
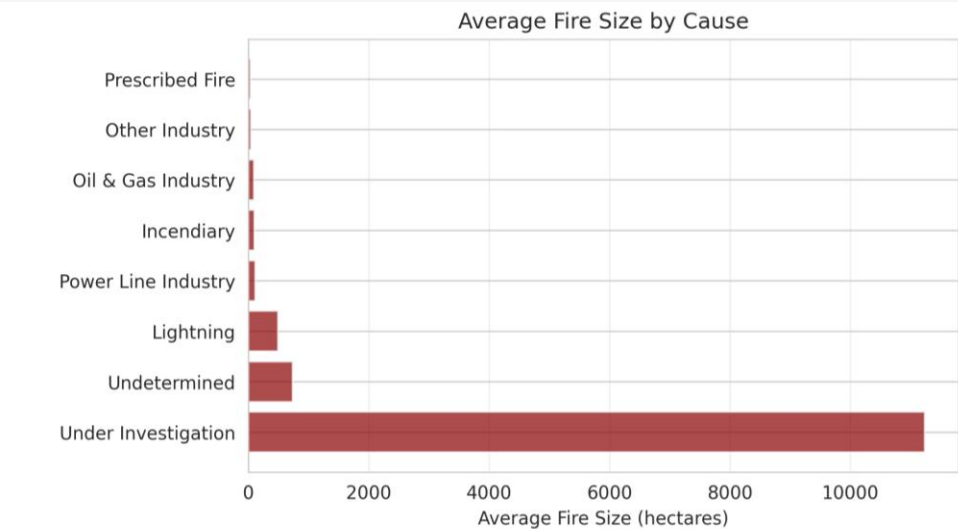
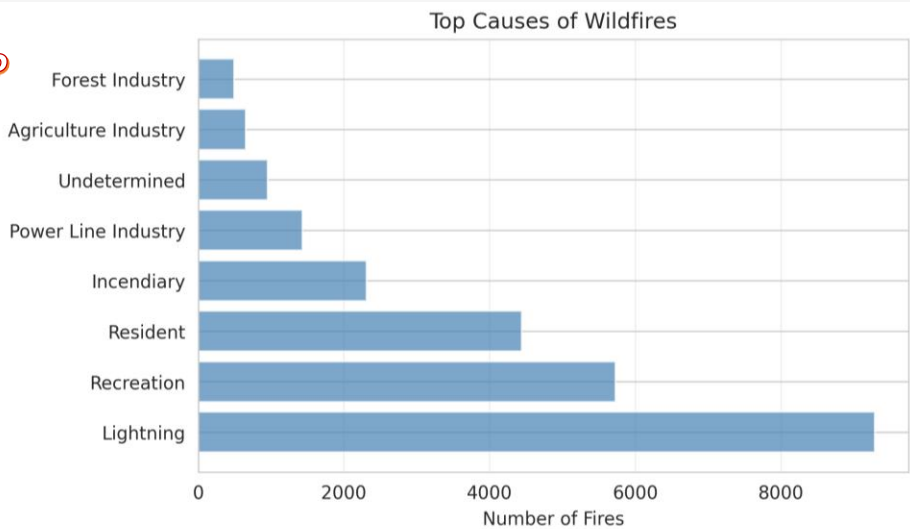
What ignites fires, and where?



STRATEGY IMPLICATIONS:

- **North:** Invest in lightning detection networks
- **South:** Public education and prevention programs

Chi-Square Test: $\chi^2 = 3,847$, $p < 0.001$ → Causes vary significantly by region



Q4: Response Effectiveness

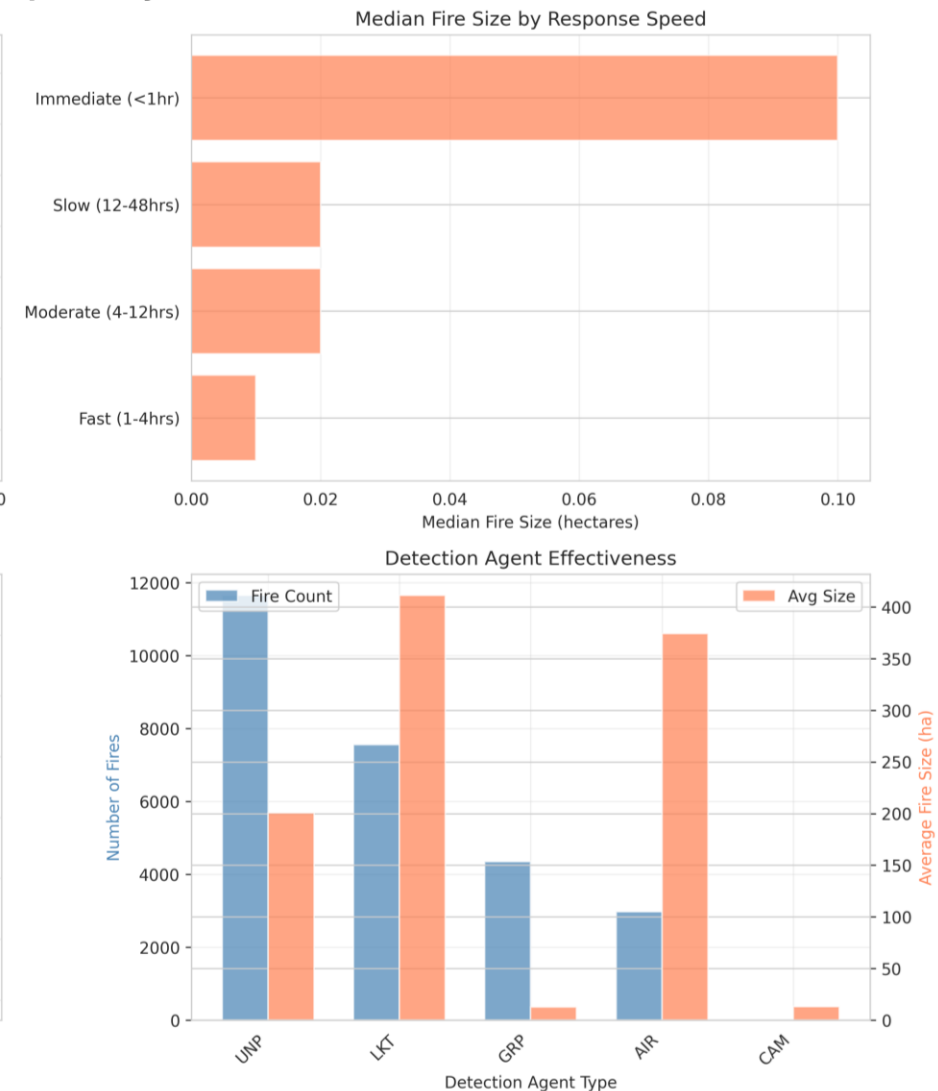
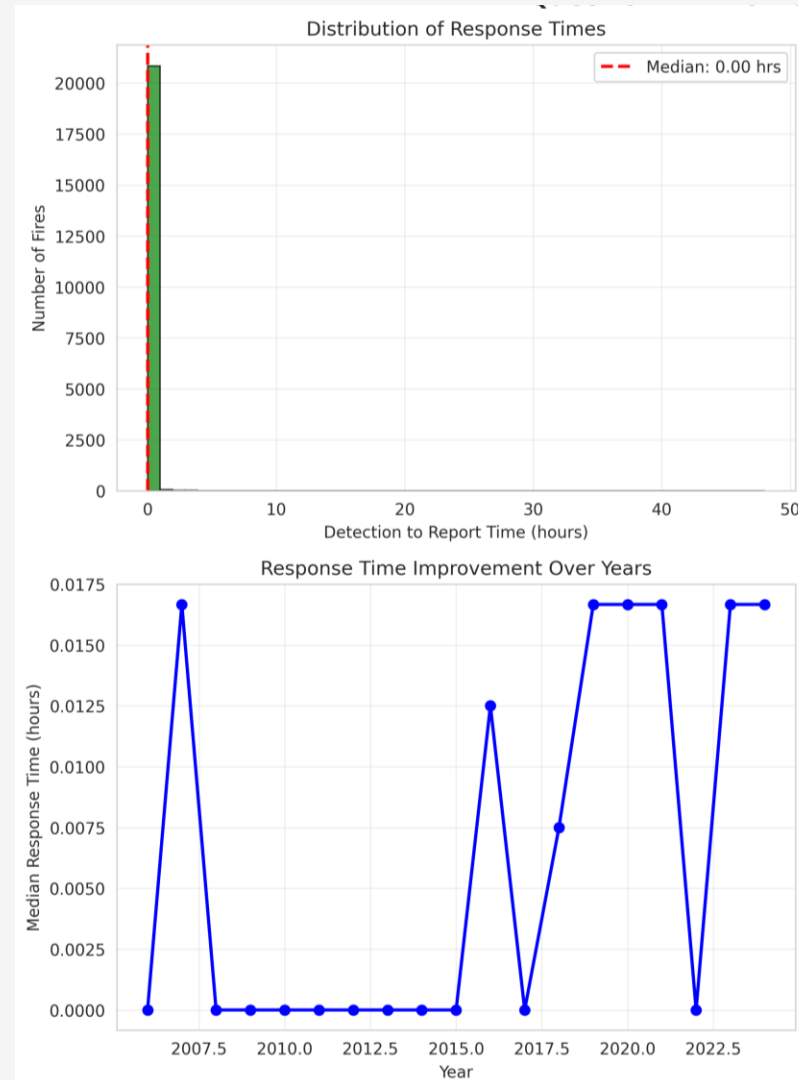
Does faster detection reduce fire size?

Weak correlation ($r \approx 0.3$)
Response time alone doesn't guarantee small fires

WHY THE WEAK RELATIONSHIP?

- Fast-growing fires trigger faster response
- Remote fires are slow to reach AND dangerous
- Suppression effectiveness varies widely
- Weather can override response speed

KEY TAKEAWAY
Invest in **BOTH**
speed AND
prediction



Q5: Environmental Factors

Which weather factors predict fire behavior?

Temperature
 $r = 0.030$

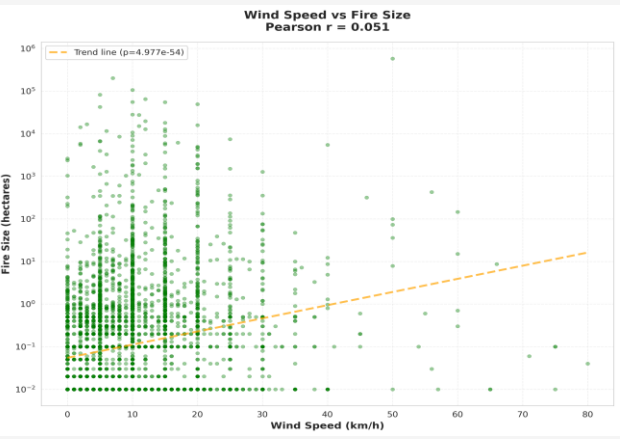
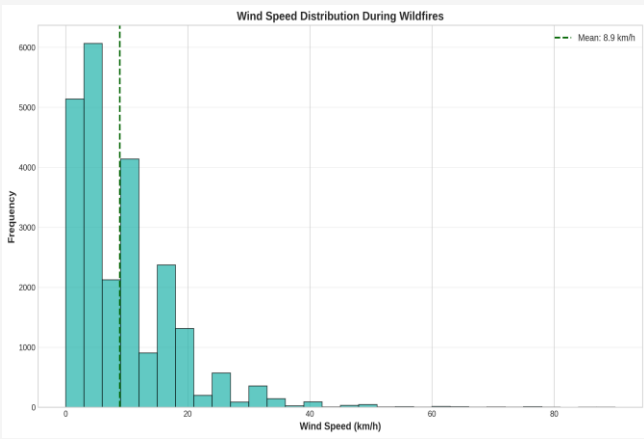
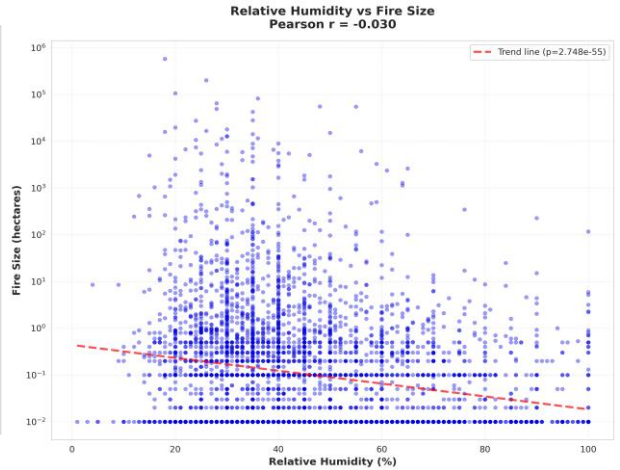
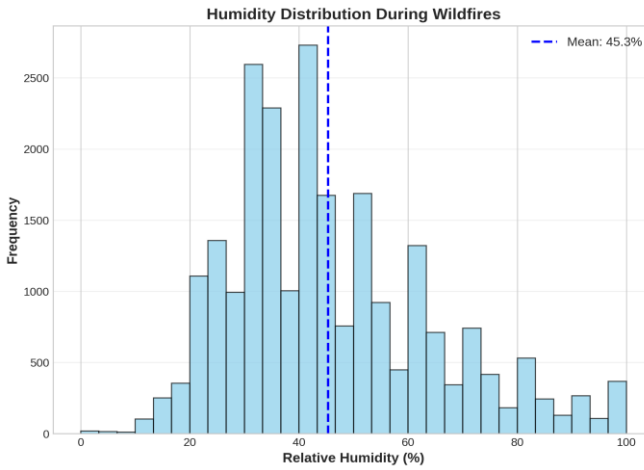
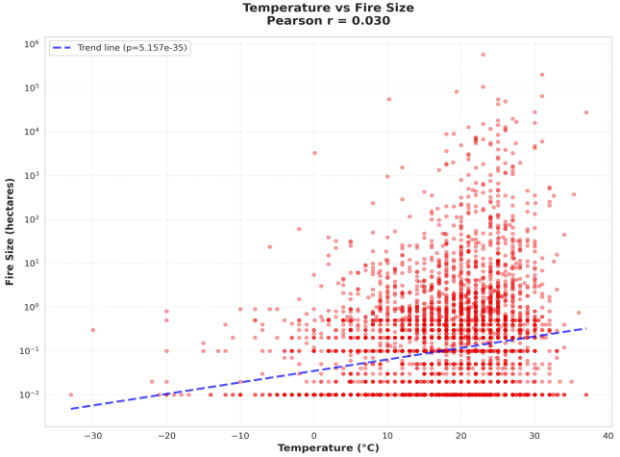
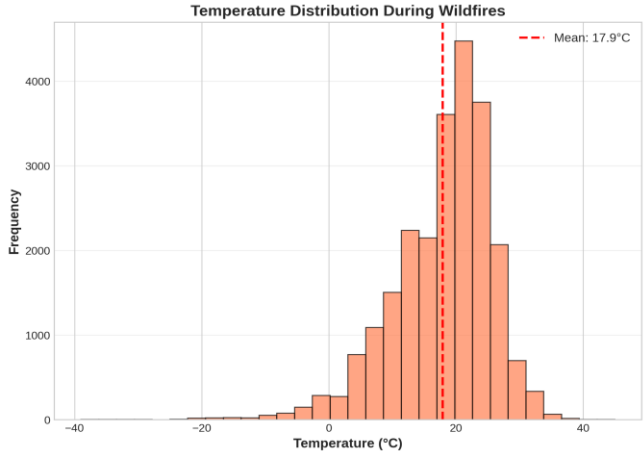
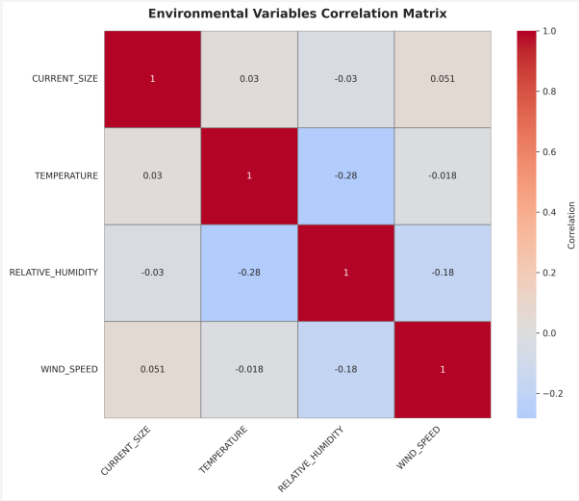
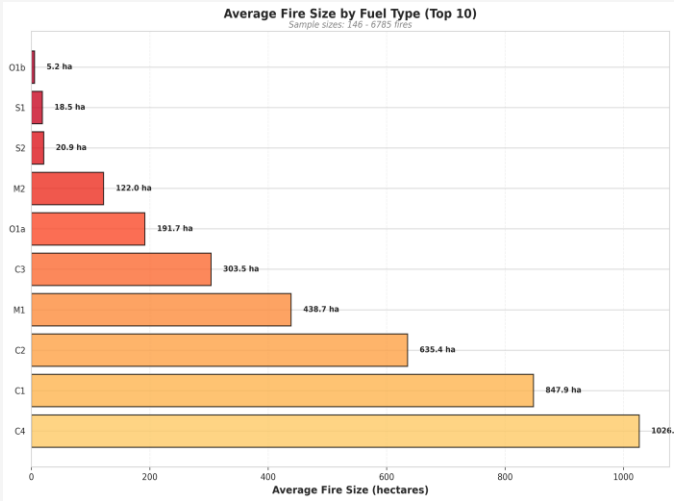
Humidity
 $r = -0.030$

Wind Speed
 $r = 0.051$

CRITICAL INSIGHT
Combinations
matter most

Hot + Dry + Windy
= Extreme Risk

- Higher temperatures and lower humidity are associated with larger fires.
- Fire size increases by approximately 2.8x from coolest to warmest conditions.



Q6: MACHINE LEARNING MODELING

Can we identify fire types and predict severity?

UNSUPERVISED K-MEANS CLUSTERING

4 fire behaviour types/Regimes discovered

- Cool-Humid Suppressed Regime
- Explosive Spread Regime
- Wind-Driven Regime
- Warm-Dry Baseline Regime

- Unlike fire causes (which vary by region), fire behaviour regimes occur province-wide.
- Any region can experience explosive spread conditions under the right weather.
- This means all regions need surge capacity for extreme events, not just the north.

Regional Patterns

- **Cluster 1:** Scattered across all regions but sparse.
- **Cluster 2:** Appears concentrated in certain hotspots (northern central).
- **Cluster 3:** Widespread, particularly in central/southern.
- **Cluster 0:** Distributed throughout.

Principal Component Analysis

- ❑ PC1 captures 36.2% of the variance
- ❑ PC2 captures 25.6% = 61.8% total variance explained in 2D.
- **Cluster 0 (Blue):** Large, compact group on the right—most common fire type.
- **Cluster 1 (Orange):** Extreme outliers on the left—very different fire behaviour.
- **Clusters 2 & 3:** Overlap in the middle (share some characteristics)

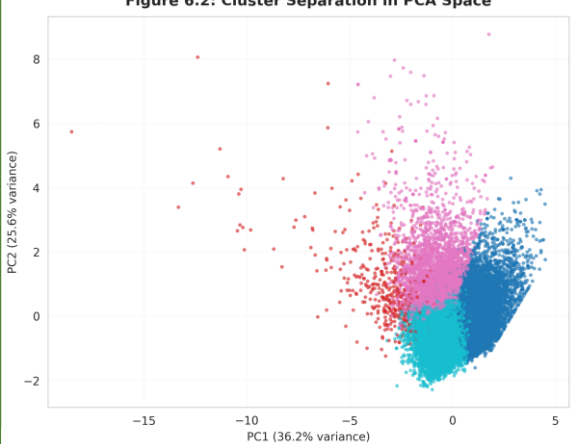


Figure 6.2: Cluster Separation in PCA Space

- Weather drives regimes more than location
- The 4 fire types are statistically distinct patterns, not arbitrary groupings.
- The 62% variance captured suggests these 2 components effectively summarize fire behaviour.

Radar Chart Visual Comparison

- Visual confirmation that Cluster 1 is the extreme danger profile.
- Clusters 2 and 3 are similar (could potentially be combined).
- Cluster 0 is the "safe" regime.

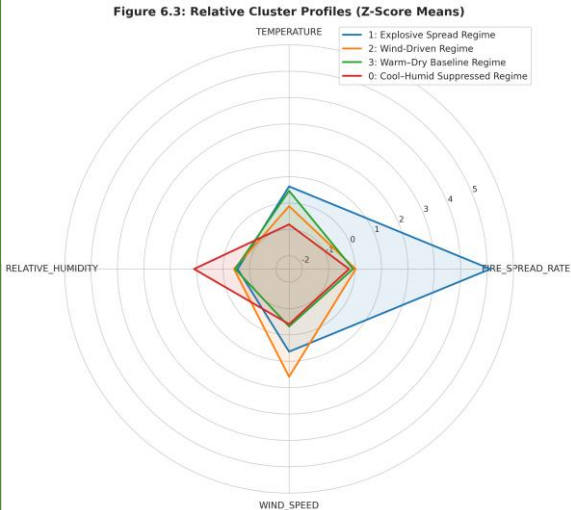
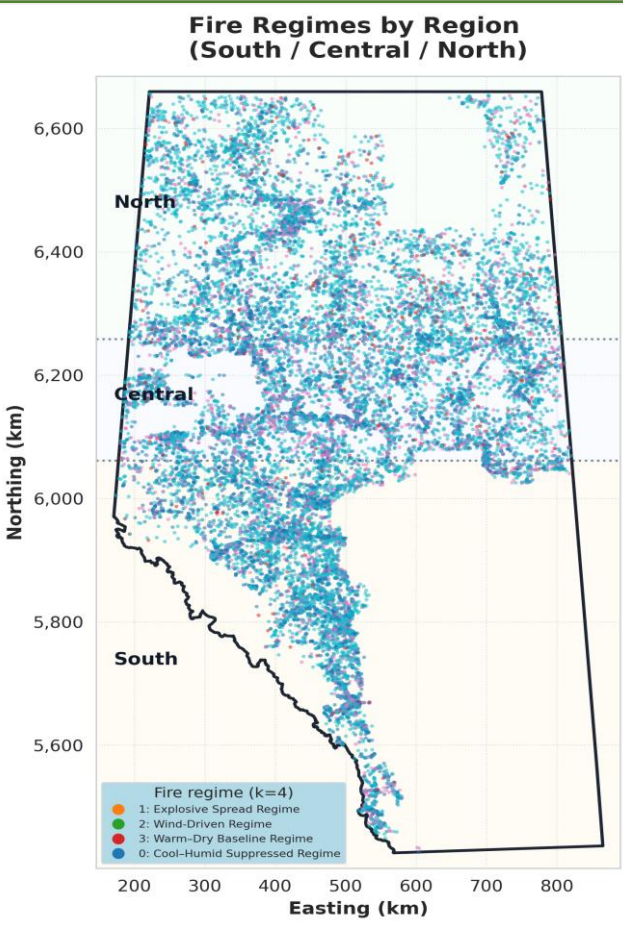
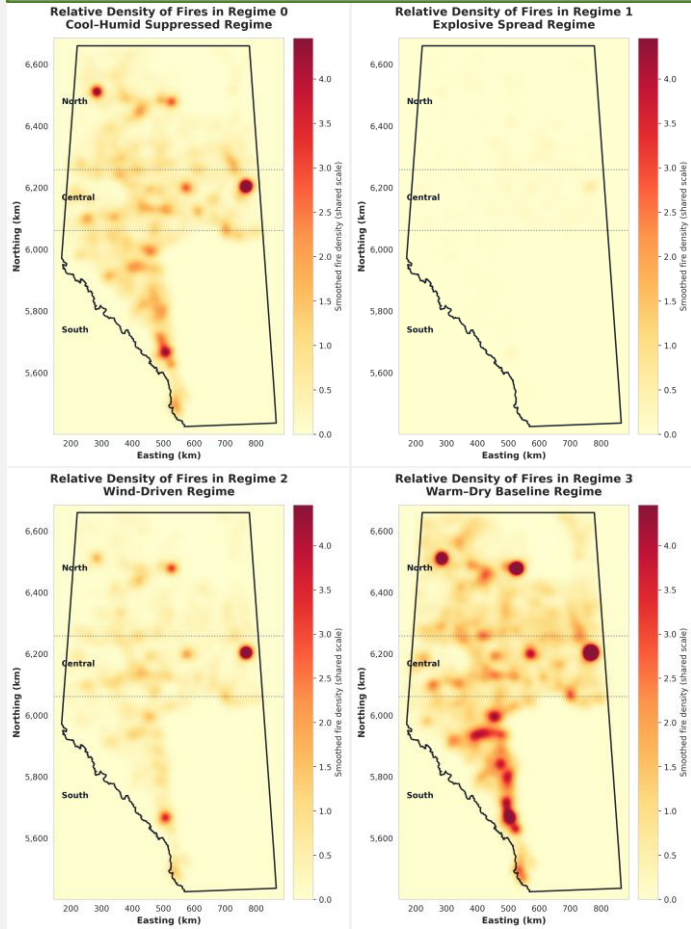
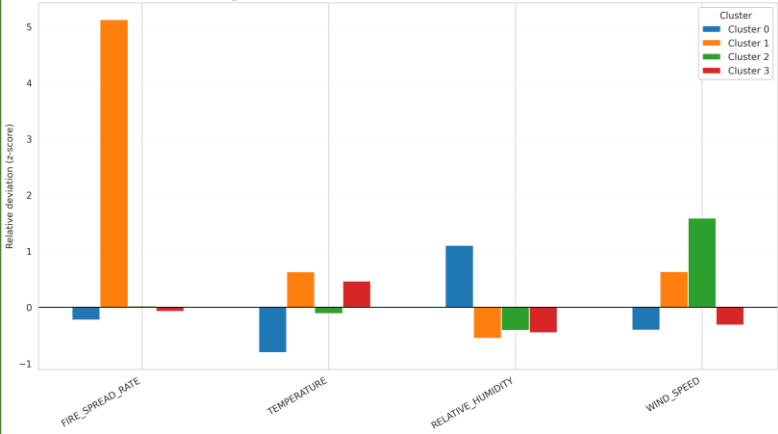


Figure 6.3: Relative Cluster Profiles (Z-Score Means)

Relative Cluster Profiles

- Cluster 1's extreme spread rate (+5 SD) makes it statistically exceptional (these are the catastrophic fire events).
- Cluster 2 has high wind but low spread, which suggests fuel type or suppression effectiveness matters more than wind alone.



Random Forest Performance

SUPERVISED RANDOM FOREST

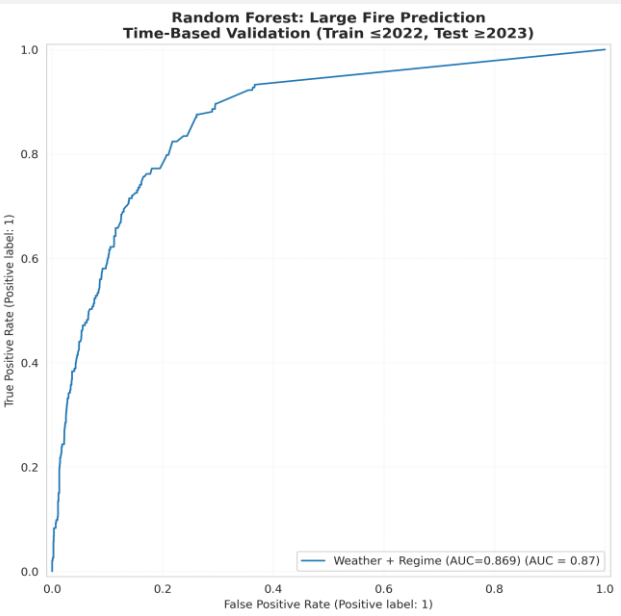
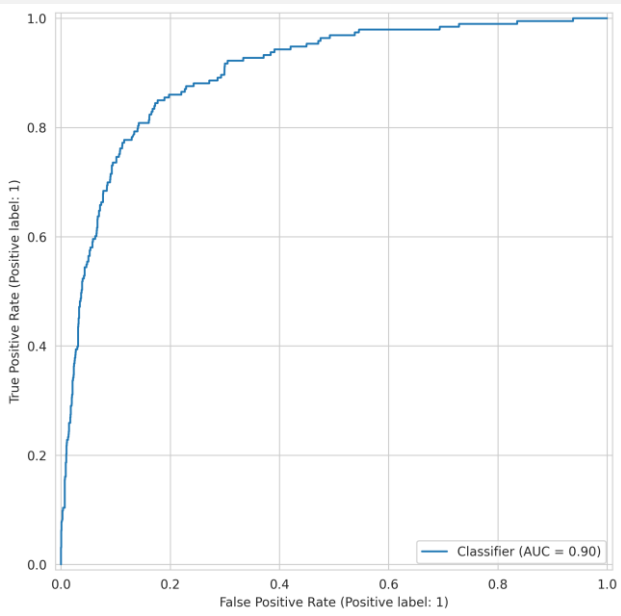
Fire Size Prediction

- ✓ 87-90% accuracy predicting size
- ✓ 81% recall (Large Fires)
- ✓ Temporal validation
- ✓ 84% Overall Accuracy

PERFORMANCE BY CLASS:

Small Fires (n=2,169): **Large Fires (n=193):**

- | | |
|------------------------|-----------------------|
| • Precision: 98.1% ✓ | • Precision: 31% ⚠ |
| • Recall: 83.9% ✓ | • Recall: 81.3% ✓ |
| • F1-Score: 0.904 ✓ | • F1-Score: 0.449 |
| • Class Imbalance: 92% | • Class Imbalance: 8% |



OPERATIONAL FRAMEWORK

Combining Clustering + Classification

New Fire Detected

Weather Conditions Measured

Regime 1
(Explosive)
+
P(Large) = 0.85
(85%)

→ IMMEDIATE FULL
SUPPRESSION

Regime 3
(Warm-Dry)
+
P(Large) = 0.45
(45%)

→ ENHANCED
MONITORING

Regime 0
(Suppressed)
+
P(Large) = 0.10
(10%)

→ STANDARD
INITIAL ATTACK

TEMPORAL VALIDATION

Does the model work on future data?

TRAINING DATA

2006-2022
(17 years)

TEST DATA

2023+
(Future data)

TEMPORAL AUC = 0.87

MODEL LIMITATION & TRADEOFFS

Honest assessment of performance

⚠ **THE PRECISION PROBLEM**

2 out of 3 'large fire' predictions are FALSE ALARMS
(31% precision)

Cost of False Positives

- Wasted resources
- Unnecessary deployment
- 69% false alarm rate

Cost of False Negatives

- Missed large fires
- Delayed response
- 19% miss rate ⚠

OPERATIONAL
DECISION:

→ Accept false
alarms to catch
dangerous fires

KEY FINDINGS:

- ✓ No temporal degradation - patterns stable across years
- ✓ Fire behavior predictable from weather variables
- ✓ Adding regime clusters doesn't improve AUC beyond weather alone
- Regimes provide interpretability, not additional prediction power

Synthesis: Five Key Insights

1. Patterns Are NOT Random

Geographic clustering + distinct types + weather correlations

2. High year-to-year variability with no simple linear trend

3. Three Distinct Fire Environments

Northern (remote/large) | Central (mixed) | Southern (human/small)

4. Weather Combinations Matter

Hot + Dry + Windy = Extreme risk (not individual factors)

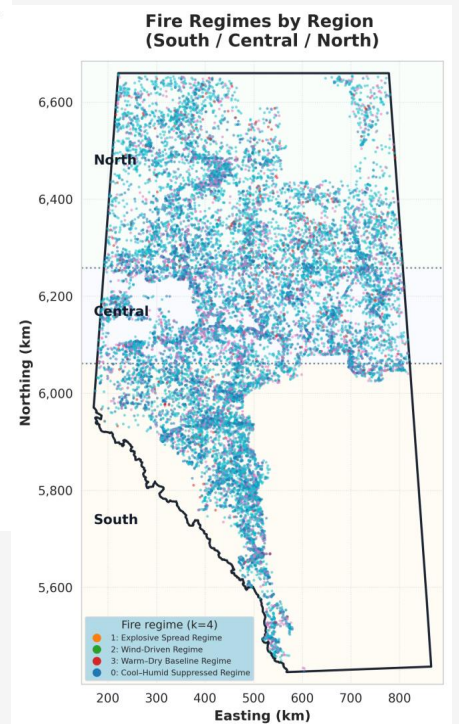
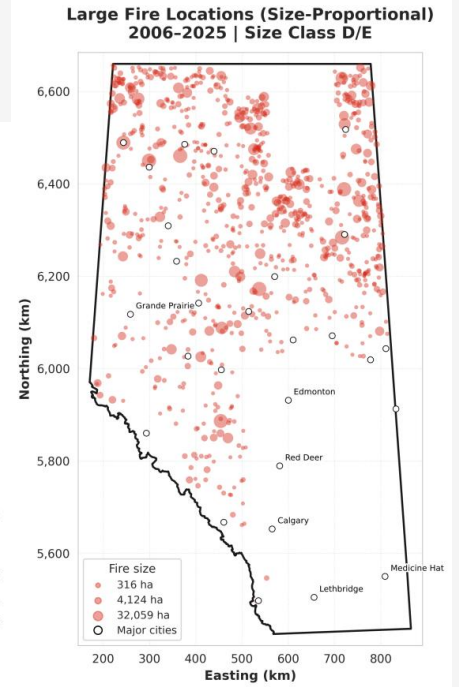
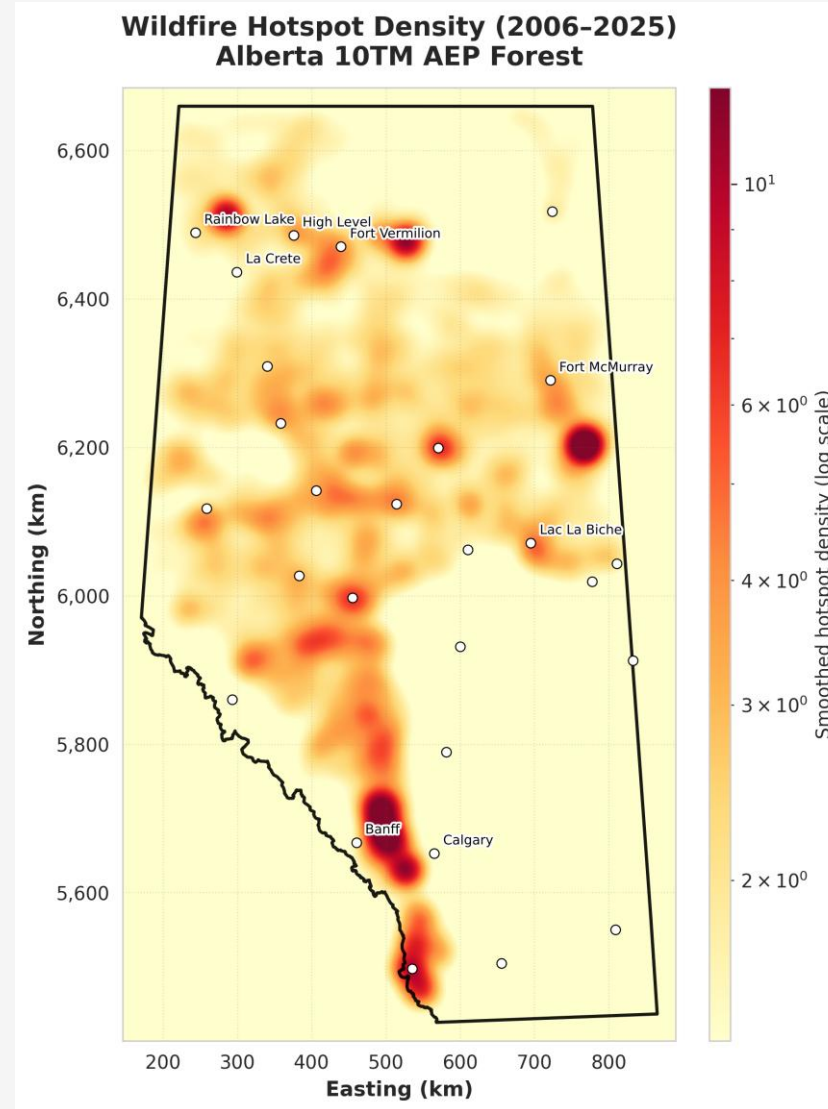
5. Speed ≠ Success

Weak correlation ($r \approx 0.03$) → Invest in BOTH speed AND prediction

6. ML Shows Promise

87% accuracy but 13% error → Supports experts, doesn't replace

7. Regional strategies outperform province-wide approaches



From Data to Decisions

Can 20 years of data improve fire management?

YES, but with nuance

FOR MANAGERS:

- ✓ Use cluster profiles for risk assessment
- ✓ Pre-position resources by geography
- ✓ Implement regional strategies

FOR POLICY:

- ✓ Regional > province-wide strategies
- ✓ North: detection | South: prevention
- ✓ Prepare for high variability

*Data cannot replace expertise.
But data CAN empower professionals with evidence.*

Thank You

26,551 fires analyzed • 6 questions answered • 4 fire types discovered

Alberta Wildfire Data Story (2006-2025)

Data Science Portfolio Project • February 2026